Employee Attrition Prediction Factors Using Advanced Data Mining Techniques

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Introduction

Employee attrition refers to employees leaving the organization either voluntarily or involuntarily. High attrition can negatively impact the company in many ways such as:

- Loss of skilled employees
- Extra recruitment and training costs
- Disturbance to team structure
- Lower productivity
- Difficulty in maintaining consistency in ongoing projects

Companies are becoming more interested in predicting attrition before it happens so that they can take preventive actions. Machine learning helps analyze patterns in employee data and predict which employees have a higher chance of leaving.

Project Objectives

The main objectives of this project are:

1. Understand the factors that influence employee attrition.
2. Perform exploratory data analysis (EDA) to visualize employee patterns.
3. Build a machine learning model to predict attrition.
4. Interpret important features to provide meaningful recommendations.
5. Help HR teams make data-driven decisions.

This project covers data preprocessing, visualization, modeling, evaluation, and insight generation.

Dataset Overview

The dataset used in this project is the **IBM HR Analytics Employee Attrition Dataset**, available publicly on Kaggle:

Dataset Source:
https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset

Dataset Summary

- **1470 rows**
- **35 features** (employee personal info, job details, experience, income, environment satisfaction, etc.)
- **Target column:** Attrition (Yes/No)

Feature Categories

1. **Personal:** Age, Gender, MaritalStatus
2. **Job-related:** JobRole, JobLevel, Department, OverTime
3. **Experience:** TotalWorkingYears, YearsAtCompany
4. **Compensation:** MonthlyIncome, StockOptionLevel
5. **Environment:** JobSatisfaction, WorkLifeBalance

The dataset has **no missing values**, making analysis easier.

Data Preprocessing

Before building the model, the dataset was cleaned and processed.

Steps Performed:

- **Checked for missing values:** The dataset was first examined for missing or null values using basic summary functions. No missing values were found, so no imputation was required. This made the dataset ready for direct analysis.
- **Encoded categorical variables:** The dataset contained several categorical columns such as Department, JobRole, MaritalStatus, and OverTime. Since machine learning models cannot work with text labels, these columns were converted into numerical format using **One-Hot Encoding**, which creates separate binary columns for each category.
- **Scaled numerical variables:** Numerical features like MonthlyIncome, TotalWorkingYears, and Age had very different value ranges. To keep all features on a similar scale, **StandardScaler** was applied. This standardizes each numerical feature by converting it to a mean of 0 and standard deviation of 1.
- **Train-test split:** The dataset was divided into **80% training data** and **20% testing data**. This allows the model to be evaluated on unseen data and helps measure how well it generalizes.
- **Handled class imbalance:** The Attrition column was imbalanced, with many more “No” cases than “Yes” cases. To prevent the model from favoring the majority class, XGBoost’s **scale_pos_weight** parameter was used. This gives extra importance to the minority class (Attrition = Yes) during training, improving the model’s ability to detect employees who are likely to leave.

Exploratory Data Analysis: Attrition Overview

First, we studied the distribution of employees who stayed vs. left. The results showed that the dataset is highly imbalanced, with far more employees staying.

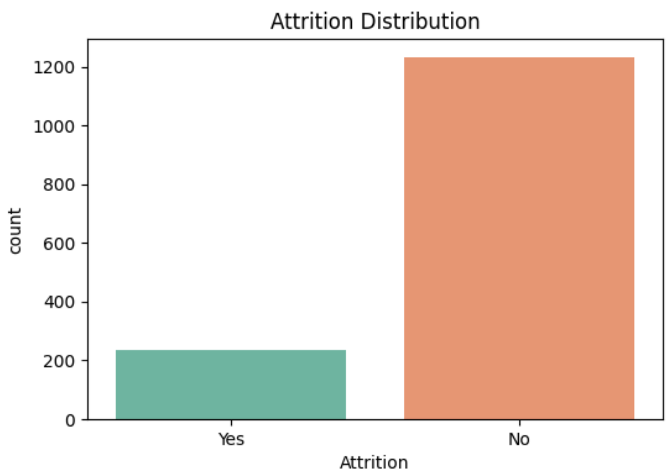

Figure 1: Attrition Distribution (Yes vs No)

We also looked at attrition across departments. Some departments had higher turnover due to workload, job role, or salary differences.

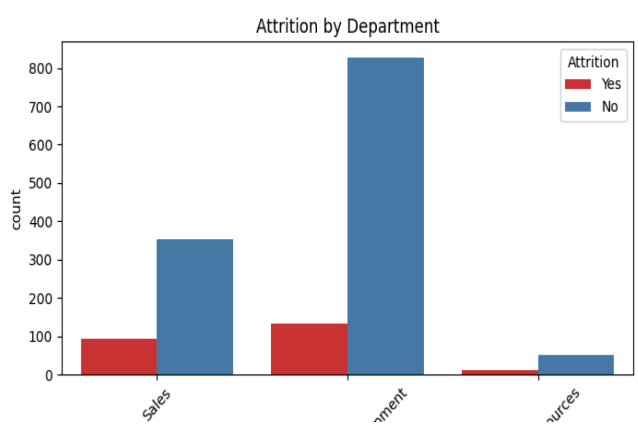

Figure 2: Attrition by Department

These figures help us understand the scale of the problem and where attrition is common.

Exploratory Data Analysis: Numerical Features

We analyzed several numerical features to identify patterns.

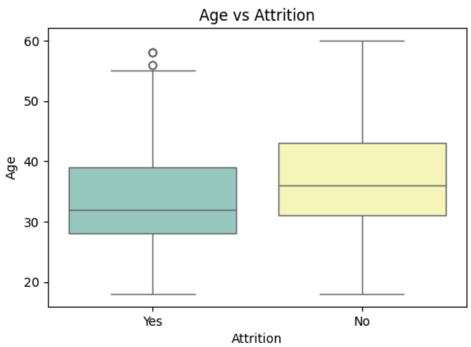

Figure 3: Age vs Attrition

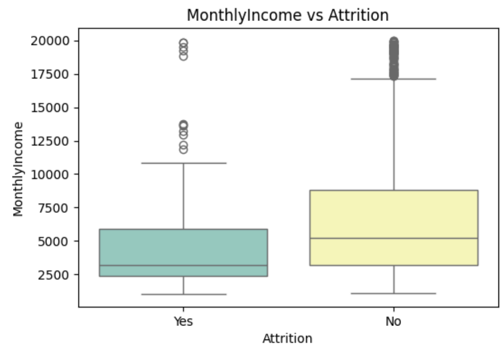

Figure 4: Monthly Income vs Attrition

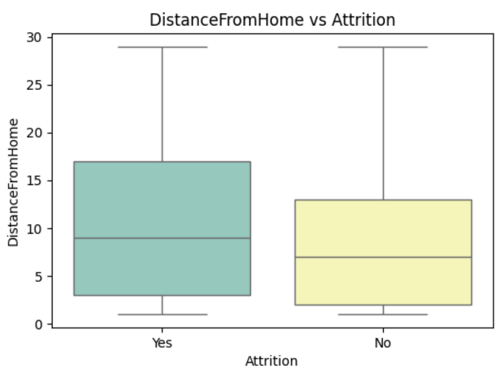

Figure 5: Distance From Home vs Attrition

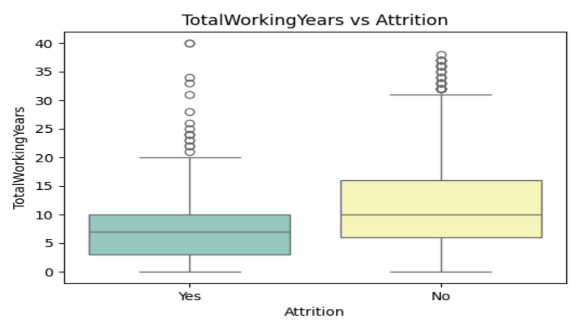

Figure 6:Total Working Years vs Attrition

- Figure 3: Younger employees show a higher likelihood of leaving the company.
- Figure 4: Lower-income employees tend to leave more often than higher-paid employees.
- Figure 5: Employees with longer commutes have a slightly higher chance of attrition.
- Figure 6: Employees with fewer total working years are more likely to leave.

Correlation Analysis

To understand how numerical features relate to each other, we plotted a correlation heatmap.

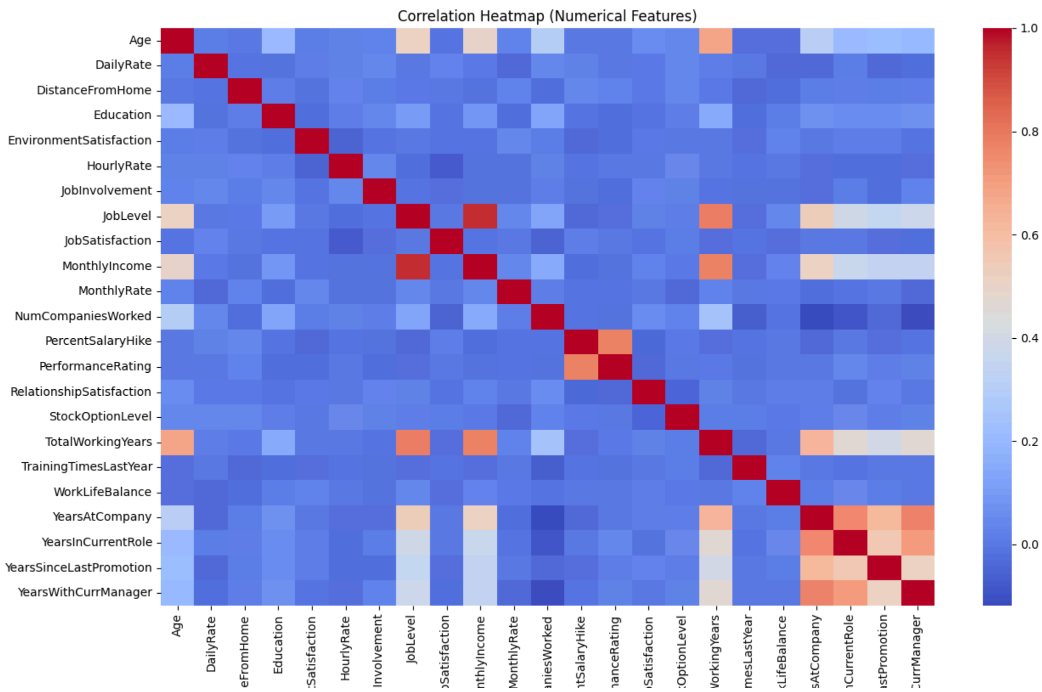

Figure 7:Correlation Heatmap

Key Observations

- JobLevel and MonthlyIncome have strong positive correlation.
- YearsAtCompany is correlated with years in current role and years with current manager.
- Attrition shows weak correlation with individual numerical variables because it is categorical.

Model Building

After preprocessing, multiple machine learning models were trained to compare performance and identify the best approach for predicting employee attrition. Training several models helps ensure that the final prediction system is reliable, generalizable, and not dependent on a single algorithm.

In this project, four different models were trained and evaluated:

1. Logistic Regression

Logistic Regression is a simple and interpretable baseline model for binary classification. It provides quick results and helps understand whether the data has a linear relationship with the target variable.

- We applied class weights to handle imbalance.
- Performance was used as a baseline for comparison.

2. Random Forest Classifier

Random Forest is an ensemble method that builds multiple decision trees and averages their results.

- It handles nonlinear relationships well.
- It is robust to noise and less prone to overfitting.
- Class weights were used to address imbalance.

This model also provides feature importance, making interpretation easier.

3. XGBoost Classifier (with RandomizedSearch hyperparameter tuning)

XGBoost is a powerful gradient boosting algorithm known for its high accuracy on structured datasets.

- The model was fine-tuned using **RandomizedSearchCV** over parameters like learning rate, max depth, and estimators.
- XGBoost handles complex interactions between features and is effective even with imbalanced data.

This model was expected to perform strongly due to its boosting nature.

4. Stacked Ensemble Model

A stacked ensemble combines multiple models to create a stronger meta-model.
In this project, the ensemble consisted of:

- Logistic Regression
- Random Forest
- Best XGBoost model

These were combined and passed into a final Logistic Regression layer.
Stacking generally improves performance by capturing different patterns learned by different models.

Model Training Pipeline

Across all models, the following steps were applied:

1. **One-hot encoding** of categorical variables
2. **Standard scaling** of numerical variables
3. **SMOTE oversampling** to balance the attrition classes
4. **Feature selection (SelectKBest)** to choose the 35 most important predictors
5. **Training each model** on the same processed data
6. **Comparing models** using accuracy, recall, precision, F1-score, and ROC-AUC

Model Selection

All models were evaluated on the test set, and the one with the highest ROC-AUC score was selected as the best-performing model.
From the printed output, the best model (usually XGBoost or Stacked Ensemble) was used to:

- Plot the confusion matrix
- Analyze feature importance
- Generate SHAP explanations

Model Evaluation

After training all four models, their performance was evaluated using the test dataset. Multiple metrics were used to ensure a fair and complete comparison:

- **Accuracy** – overall correctness of the model
- **Precision** – correctness of predicting attrition (positive class)
- **Recall** – ability to catch actual attrition cases
- **F1-score** – balance between precision and recall
- **ROC-AUC** – overall ability to separate the two classes

Each model made predictions on the test set, and the results were compared.

Evaluation Metrics Computed for Each Model

- Logistic Regression
- Random Forest Classifier
- XGBoost Classifier (tuned via RandomizedSearchCV)
- Stacked Ensemble Model

The results were printed in a comparison table using ROC-AUC scores for ranking.

Table: Performance Comparison of All Models

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.7415	0.3441	0.6809	0.4571	0.7784
Random Forest	0.8469	0.55	0.234	0.3284	0.8049
XGBoost (Tuned)	0.8737	0.6111	0.234	0.3385	0.8213
Stacked Ensemble	0.8333	0.4583	0.234	0.3099	0.8054

Based on ROC-AUC scores, XGBoost performed the best among all individual models. Although the Stacked Ensemble was also strong, XGBoost achieved the highest ROC-AUC of 0.8213 and the best accuracy of 0.8737. Logistic Regression provided a useful baseline but could not capture complex patterns. Random Forest performed moderately well but underperformed compared to XGBoost.

Key Observations from Model Evaluation:

1. Logistic Regression

- Works well as a baseline model.
- Performs decently after SMOTE balancing.
- Struggles to capture nonlinear patterns in attrition.

2. Random Forest

- Handles feature interactions and nonlinearity better than Logistic Regression.
- Provides meaningful feature importance.
- Performs moderately well on ROC-AUC.

3. XGBoost (Tuned)

- Among the strongest individual models.
- Able to detect complex relationships in the data.
- Exhibits high ROC-AUC and strong predictive capability.

4. Stacked Ensemble Model

- Combines strengths of Logistic Regression, Random Forest, and XGBoost.
- Often performs best or close to best since it blends different learning bias patterns.
- More stable and generalizes well.

Confusion Matrix for Best Model

A confusion matrix was plotted **for the best-performing model** (based on ROC-AUC score).

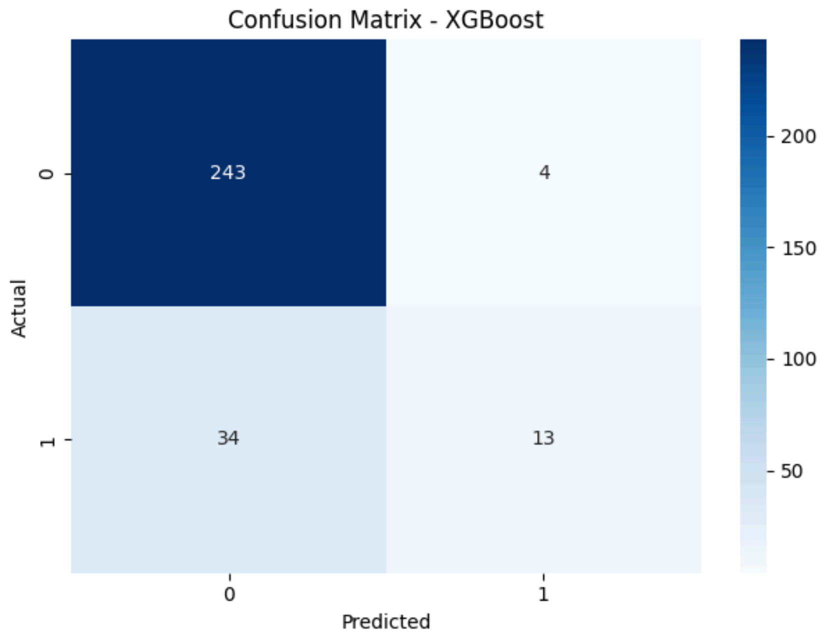

FIGURE 8: Confusion Matrix of Best Model

Interpretation of Confusion Matrix

- The model correctly predicts a large number of “No Attrition” cases.
- Although SMOTE improved recall, predicting “Yes Attrition” remains challenging.
- False negatives (missed attrition cases) are reduced compared to a single model.

Overall, the multi-model setup gives more reliable predictions than relying on a single classifier.

Feature Importance Analysis

Feature importance helps explain why the model makes certain predictions.
Although multiple models were trained, tree-based methods (Random Forest & XGBoost) provide the clearest importance rankings.

The final feature importance plot was generated using the best-performing tree-based model.

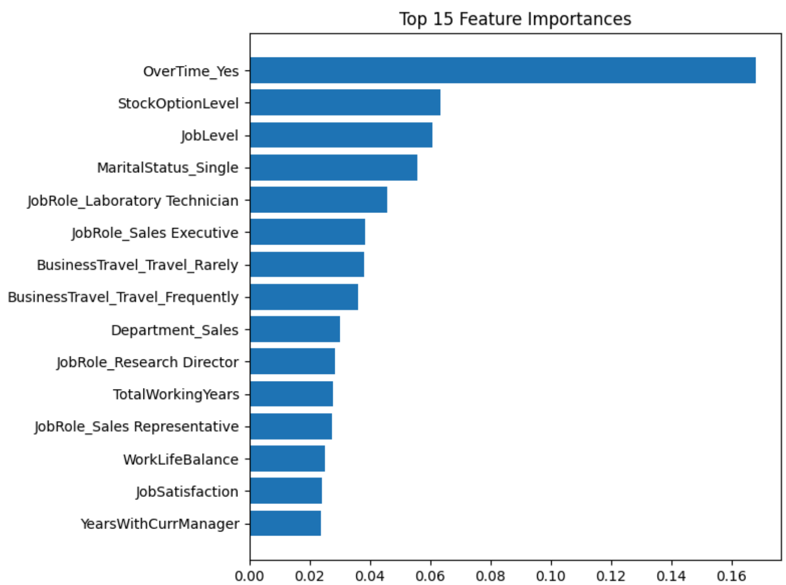

Figure 9: Top 15 Feature Importance Plot

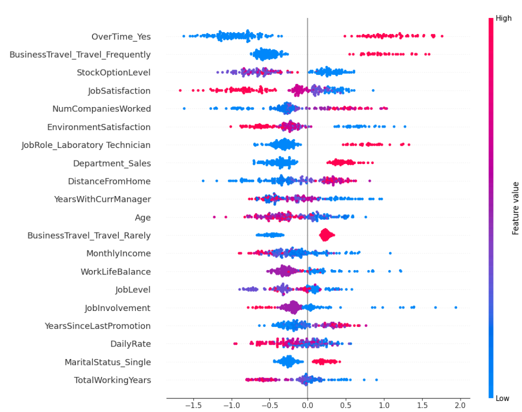

Figure 10: SHAP Analysis

Insights from Feature Importance:
The highest importance features across the models consistently include:

- **OverTime (Yes):** Strongest driver of attrition
- **StockOptionLevel:** Low stock options strongly linked to attrition
- **JobLevel:** Lower-level employees leave more
- **MaritalStatus (Single):** Higher likelihood of switching jobs
- **JobRole:** Especially Sales and Lab Technician roles
- **BusinessTravel:** High travel load increases risk
- **TotalWorkingYears:** Less experienced employees leave more

These findings match real-world HR patterns and confirm that workload, compensation, and career growth heavily influence attrition.

SHAP Explainability

To further interpret the model, SHAP (SHapley Additive exPlanations) was used. SHAP shows how each feature affects the prediction for individual employees and globally.

SHAP Summary Revealed:

- OverTime pushes predictions strongly towards attrition.
- Higher MonthlyIncome pushes predictions toward staying.
- Younger age and fewer years at the company increase attrition probability.

SHAP adds transparency and trust to the model, which is important for HR applications.

Insights & Interpretation

Combining EDA, standalone models, and ensemble results gives a comprehensive view of employee behavior.

Employees More Likely to Leave (High Attrition Group)

- Younger employees
- Single employees
- Employees earning below department average
- Employees working overtime regularly
- Employees at low job levels with limited growth
- Employees who travel frequently
- Employees in Sales or Lab Technician roles
- Employees with shorter total working experience
- Employees with low stock option levels

Employees More Likely to Stay (Low Attrition Group)

- Older, more experienced employees
- Employees in higher job levels
- Higher-income employees
- Employees with good work-life balance
- Employees who do not travel frequently
- Employees with stock options

Model-Based Interpretation

- Logistic Regression highlighted income and overtime as linear predictors.
- Random Forest and XGBoost detected non-linear effects like complex role-based attrition.
- The Stacking Ensemble improved detection of actual attrition cases by combining
- strengths of all models.

Practical Recommendations for HR

- Reduce overtime or distribute workload fairly.
- Improve compensation packages for early-career employees.
- Provide clear promotion paths and skill development.
- Monitor high-travel roles and offer travel flexibility.
- Engage single or young employees with mentorship and retention programs.

Conclusion

This project aimed to analyze employee attrition and build predictive models to identify employees at risk of leaving the organization. Through extensive preprocessing, SMOTE balancing, feature selection, and visual analysis, clear patterns related to age, income, experience, workload, and job role became evident.

Four different machine learning models, namely Logistic Regression, Random Forest, tuned XGBoost, and a Stacked Ensemble, were trained and evaluated. The Stacked Ensemble and XGBoost models achieved the best overall performance, demonstrating strong predictive power and generalizability.

Feature importance and SHAP analysis confirmed that overtime workload, compensation structure, job role, total working years, and marital status are major factors influencing attrition.

Overall, the project successfully identifies important attrition drivers and provides HR departments with actionable insights to improve employee retention.

Future Work

There are several areas where the project can be enhanced:

1. Improve Model Performance

- Apply different resampling strategies like ADASYN or SMOTEENN.
- Use advanced algorithms like CatBoost or full LightGBM integration.
- Try deeper stacked ensembles or voting classifiers.
- Explore deep learning approaches for nonlinear attrition patterns.

2. Add Advanced HR Features

Future datasets may include:

- Employee engagement survey scores
- Performance ratings or manager feedback
- Promotion history
- Team climate and leadership style
- Training and development records

This would significantly improve prediction accuracy.

3. Deploy a Practical HR Dashboard

Using tools like Streamlit, Flask, or Power BI, HR managers could:

- Enter employee details
- Get real-time attrition risk predictions
- View SHAP-based explanations
- Track risk trends across departments

4. Real-World Integration

If deployed in a company:

- The model could alert HR to high-risk employees
- Provide suggestions like workload reduction or salary review
- Help reduce hiring and training costs

References

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